

# Generalization. Out of Sample Performance.

## Big Data y Machine Learning para Economía Aplicada

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# Agenda

- 1 Previously ...
- 2 Generalization
  - Overfitting and Underfitting
  - Out-of-Sample Error Estimation
    - AIC: Akaike Information Criterion
    - SIC/BIC: Schwarz/Bayesian Information Criterion
  - Cross-Validation
- 3 Review

# Agenda

## 1 Previously ...

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## 3 Review

# Set up

- ▶ We observe  $(Y, X)$
- ▶ We choose a predictor  $f : \mathcal{X} \rightarrow \mathbb{R}$  to minimize expected loss.

$$f^*(X) = \operatorname{argmin}_f R(f(X)) \quad (1)$$

$$= \operatorname{argmin}_f \mathbb{E}[(Y - f(X))^2]. \quad (2)$$

# Linear model

$$f(X) = Xb \quad (3)$$

# Uncertainty

- ▶ Inference
- ▶ Prediction

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## 3 Review

# Generalization Overview

- ▶ In ML we care in out-of-sample prediction
- ▶ Generalization refers to a model's performance on unseen data.



# Generalization

- ▶ Unseen data is typically referred as **test data**,
- ▶ While the sample data is called the **training data**.
- ▶ Objective:

$$f^*(X) = \operatorname{argmin}_f \mathbb{E}[(Y - f(X))^2]. \quad (4)$$

# Linear model

Again...

$$\hat{f}(X) = X\hat{\beta} \quad (5)$$

# Linear model

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# Generalization

## Overfitting and Underfitting

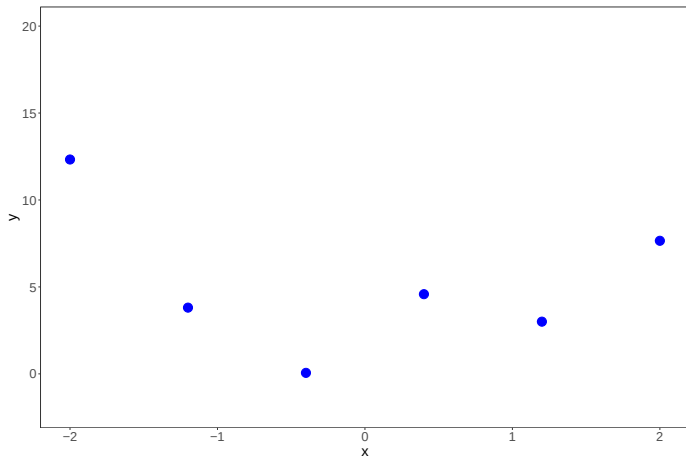


Figure 1: Training Data

# Generalization

## Overfitting and Underfitting

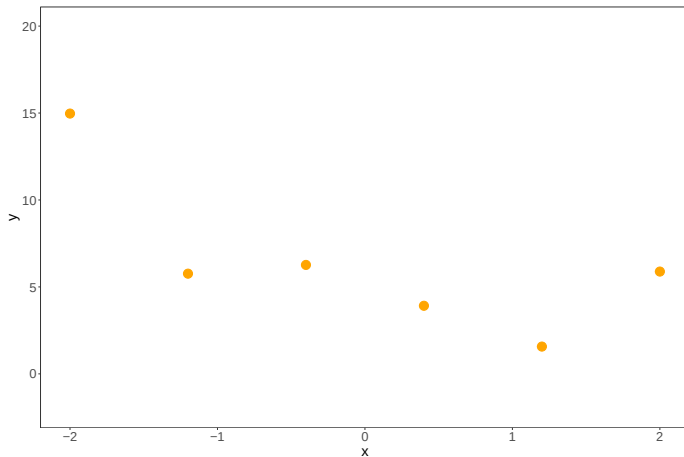
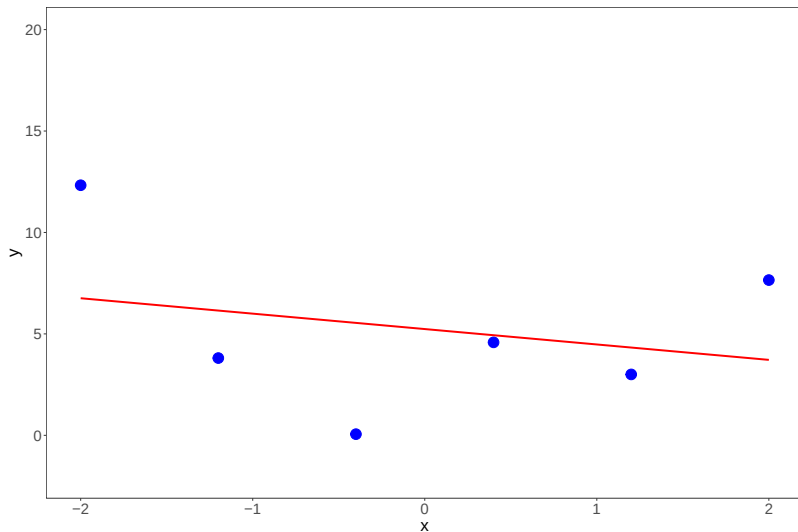


Figure 2: Testing (Unseen) Data

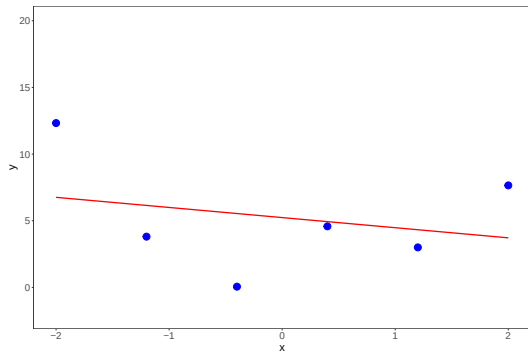
# Generalization

## Overfitting and Underfitting

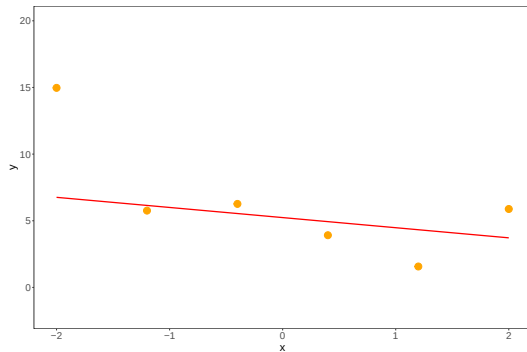


# Generalization

## Out-of-Sample Performance



(a) Training Data

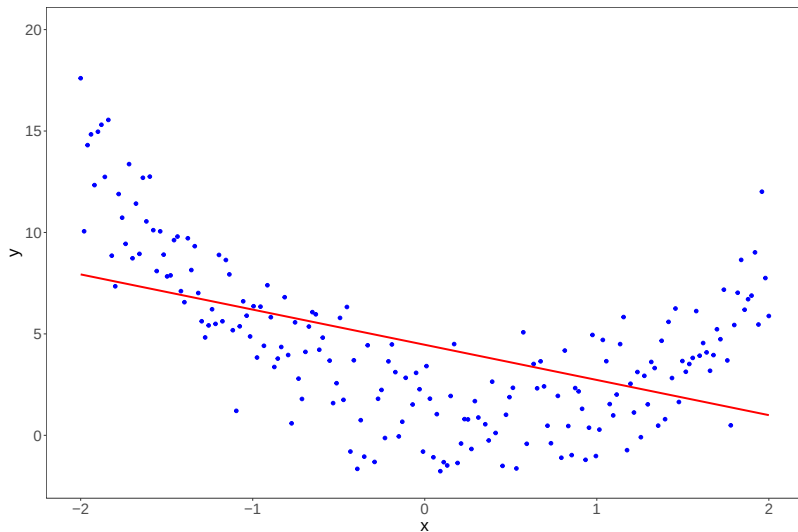


(b) Testing Data



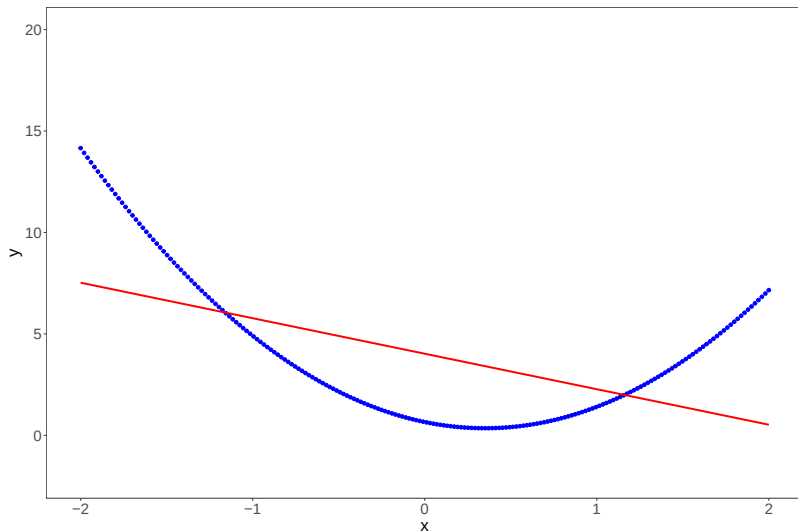
# Generalization

Out-of-Sample Performance. More data?



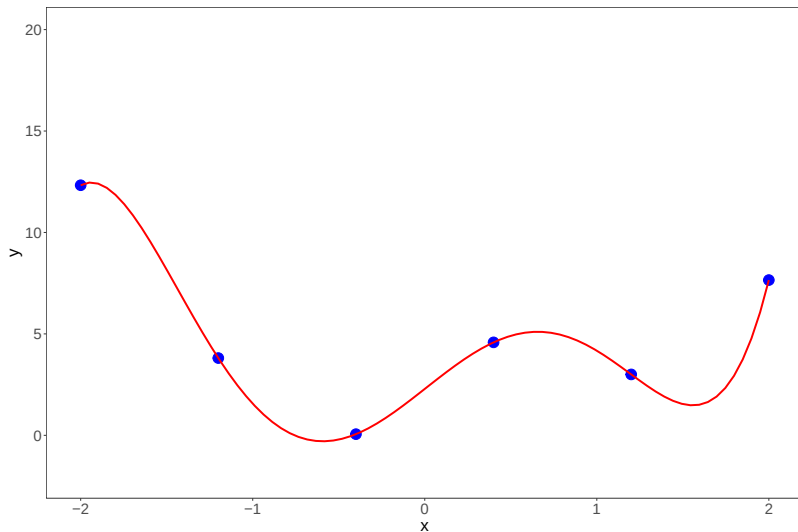
# Generalization

Out-of-Sample Performance. Noiseless data?



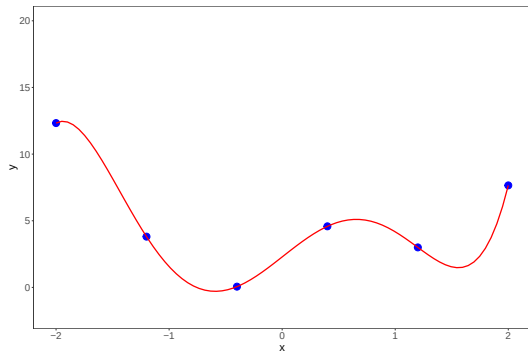
# Generalization

## More Complex Model

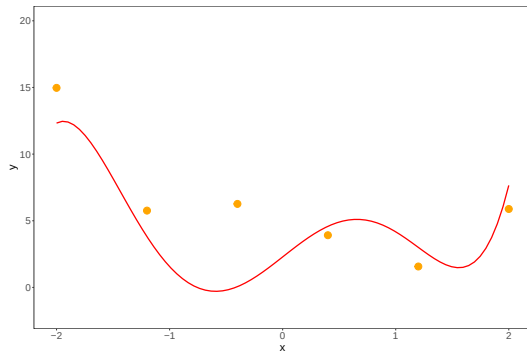


# Generalization

More Complex Model. Out-of-Sample Performance



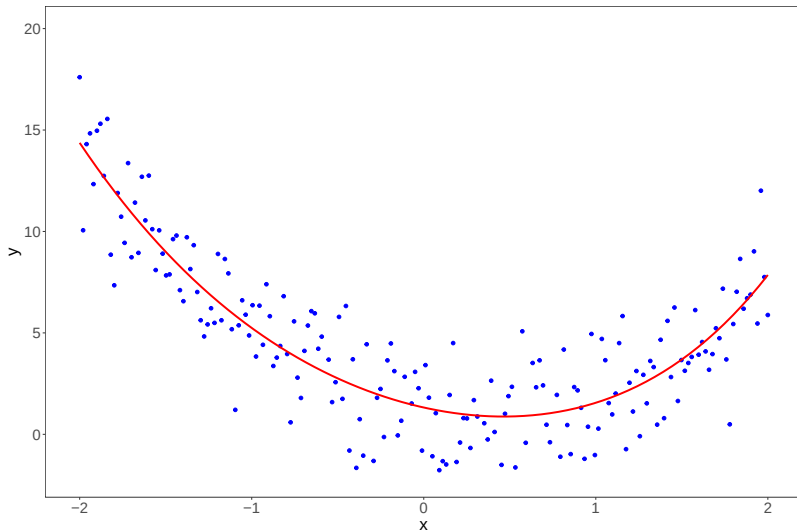
(a) Training Data



(b) Testing Data

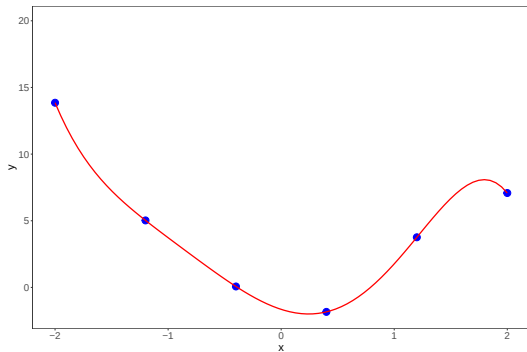
# Generalization

More Complex Model. Out-of-Sample Performance: More Data

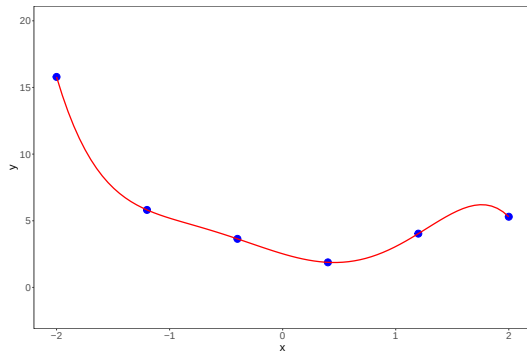


# Generalization

More Complex Model. Out-of-Sample Performance: Variance



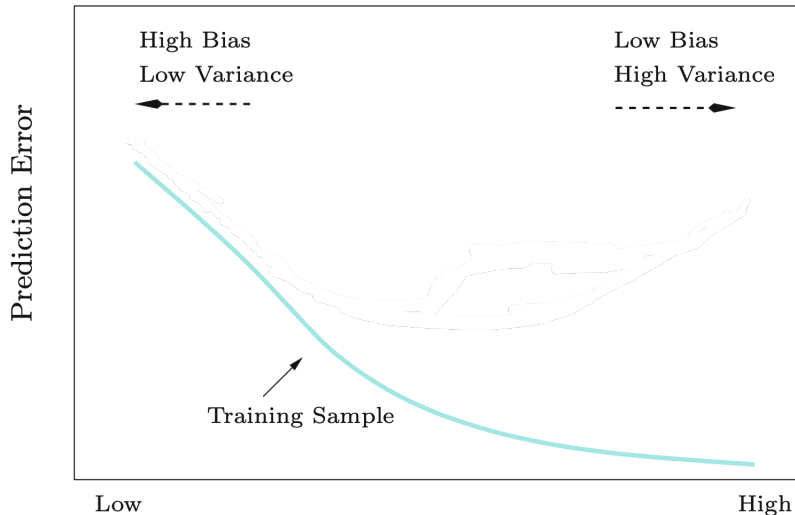
(a) Training Data 2



(b) Training Data 3

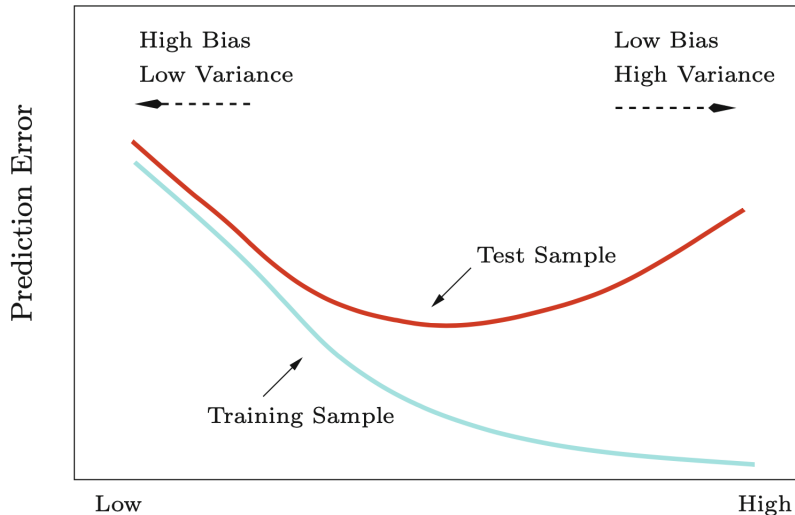
# Generalization

Overfitting and Underfitting. Bias-Variance Tradeoff



# Generalization

Overfitting and Underfitting. Bias-Variance Tradeoff

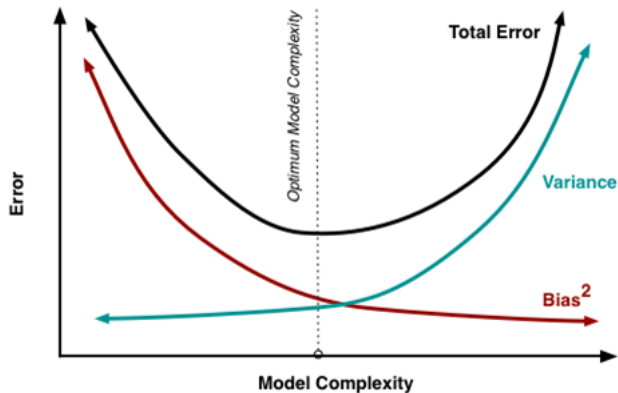




# Mathematical Decomposition for Regression

# Mathematical Decomposition for Regression

# Bias-Variance Tradeoff



Source: <https://tinyurl.com/y4lvjxpc>

- ML best kept secret: By tolerating some bias we can have significant gains in variance

# Generalization

## Key Insights on Bias-Variance Tradeoff

- ▶ The bias term reflects the error introduced by the model's inability to approximate the true function  $f^*$ .
- ▶ The variance term reflects the sensitivity of the model to the specific training set.
- ▶ As dataset size increases, variance generally decreases.
- ▶ The noise term  $\sigma^2$  is unavoidable and cannot be predicted.
- ▶ The decomposition for classification problems is less clear than for regression problems, but still present.

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# Generalization

Train and Test Sets. In-Sample and Out-of-Sample Prediction.

- ▶ Como seleccionamos la parametrización que minimize el error de predicción fuera de muestra?
- ▶ Problema: solo contamos con una muestra

# Generalization Error

- ▶ Para seleccionar la mejor parametrización con respecto al Test Error (error de prueba), es necesario estimarlo.
- ▶ Hay dos enfoques comunes:
  - ▶ Podemos estimar indirectamente el error de la prueba haciendo un ajuste al error de entrenamiento para tener en cuenta el sesgo debido al sobreajuste  $\Rightarrow$  Penalización ex post: AIC, BIC,  $MSE_{adj}$ ,  $R^2_{adj}$ , etc.

# Test Error

## AIC

- ▶ Akaike (1969) fue el primero en ofrecer un enfoque unificado al problema de la selección de modelos.
- ▶ Elegir el modelo  $j$  tal que se minimice:

$$AIC(j) = \log \left( \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2 \right) - p_j \quad (6)$$



# Test Error

## SIC/BIC

- ▶ Schwarz (1978) mostró que el AIC es inconsistente, (cuando  $n \rightarrow \infty$ , tiende a elegir un modelo demasiado grande con probabilidad positiva)
- ▶ Schwarz (1978) propuso:

$$SIC(j) = \log \left( \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2 \right) - \frac{1}{2} p_j \log(n) \quad (7)$$

# Test Error

## AIC vs BIC

$$AIC(j) = \log \left( \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2 \right) - p_j \quad (8)$$

$$SIC(j) = \log \left( \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2 \right) - p_j \frac{1}{2} \log(n) \quad (9)$$

# Test Error

$$MSE_{adj}, R^2_{adj}$$

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$$MSE_{adj}, R^2_{adj}$$

# Generalization Error

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  - ▶ Usar la misma muestra y estimar directamente el Test Error

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## Cross-Validation



photo from <https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/>

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# Review

Hoy

- ▶ Bias-Variance Tradeoff (Dilema Sesgo/Varianza)
- ▶ Sobreajuste y Selección de modelos
  - ▶ AIC, BIC,  $R^2_{adj}$
  - ▶ Enfoque de Validación
  - ▶ LOOCV
  - ▶ K-fold Cross-Validation (Validación Cruzada)